ABSTRACT

The current job market characterized by fierce competition and fast-paced technological changes makes it hard for the job seekers to figure out the precise skills needed for the positions they want. Old-style career counseling and resume reading methods are heavily dependent on key words matching and human assessment that usually do not gauge the true skills of the applicant. The **AI-Based Skill Gap Analyzer and Personalized Learning Recommendation System** is aimed at tackling the aforementioned problem by facilitating the skill evaluation and career guidance through the application of Artificial Intelligence and Machine Learning techniques. The system empowers the users to upload their resumes or to manually enter their skills that's will be examined through Natural Language Processing for the purpose of taking out the relevant competencies. Then these skills are compared with the ones from the industry-standard job roles to spot the lacking skills and to assess the compatibility for the specific roles. The system, on the basis of the extracted gaps, develops individualized learning paths, proposes related projects, and recommends appropriate job positions that are likely to boost employability. The methods of machine learning are giving the role recommendations much higher accuracy and there is also an administrative module that supports the good management of the users, job roles, and analytical insights in an efficient manner. The system made in Python and web-based technologies guarantees secure data handling, fast processing, and real-time interaction. By changing the conventional career assessment techniques with intelligent, AI-driven methodologies the proposed system enhances precision, cuts down the labor input, and lays out orderly career guidance for both students and professionals.